AI Clinical Triage System

Presentation

AI-Powered Clinical Triage for Hospital Reception
March 2026 — v3.0

SLIDE 1: The Vision

Every Patient Deserves Instant, Compassionate Care — In Their Own Language

Imagine walking into a hospital — anxious, in pain, possibly in a foreign city — and being greeted instantly by a warm, intelligent assistant who speaks your language, understands your symptoms, and guides you to the right doctor in under 2 minutes.

No waiting. No language barriers. No paperwork. Just care.

That's what our AI Clinical Triage System delivers — today. With real-time disease cluster scoring, 173 multilingual symptom aliases, and intelligent triage that adapts by age, gender, and clinical context.

SLIDE 2: The Problem

Hospital Reception is Broken

The Patient Experience Today

```
Patient arrives (anxious, in pain)
        ↓
Wait in queue: 15-45 minutes
        ↓
Struggle to communicate: language barriers
        ↓
```

```
Misrouted to wrong department: 30% of the time
        ↓
Wait again at wrong department
        ↓
Finally see the right doctor: 1-2 hours later
```

The Numbers Tell the Story

Problem	Impact
15-45 min average wait at reception	Patient dissatisfaction, delayed care
30% of patients misrouted	Wasted doctor time, hospital inefficiency
6-9 reception staff required for 24/7 coverage	$180K-350K/year labor cost
Zero documentation of initial complaints	Doctor has no context before consultation
Peak hours overwhelm reception	Emergency patients delayed by routine cases
Multilingual populations underserved	Pakistan alone has 4+ major languages

Who Suffers?

- **Patients** — Anxiety, delayed treatment, communication frustration
- **Hospitals** — High staffing costs, inefficient triage, liability risk
- **Doctors** — No patient context, repeated history-taking, wasted time
- **Healthcare System** — Bottleneck at the very first touchpoint

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SLIDE 3: Our Solution

AI That Talks, Listens, and Cares — In 6 Languages

What We Built

A **real-time, voice-based AI clinical triage platform** that:

1. **Greets** the patient warmly in their preferred language (pure Urdu script, no English mixing)
2. **Listens** with two-layer noise filtering — Silero VAD filters background speakers, confidence threshold drops noise artifacts
3. **Scores** symptoms in real-time against 70+ conditions using disease cluster scoring engine with 173 multilingual aliases

4. **Asks** targeted follow-up questions based on differential diagnosis — "chest pain" → asks about shortness of breath to differentiate Angina vs GERD
5. **Triages** using intelligent scoring — cross-condition boost, age/gender rules, doctor availability check
6. **Directs** the patient to the exact doctor, department, and room shown in triage scores
7. **Documents** the entire conversation + live patient screening card with printable triage slip

In Action

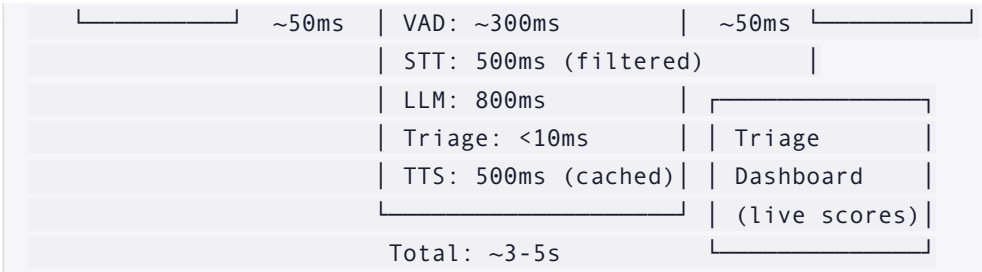
```
┌──────────────────────────────────────────────────────┐
│                                                      │
│   Patient: "مجھے سینے میں درد ہو رہا ہے"              │
│   (I'm having chest pain)                            │
│                                                      │
│   Agent (Amina): "آپ کی تکلیف سن کر افسوس ہوا۔        │
│   کب سے ہے یہ درد؟"                                  │
│   (I'm sorry to hear that. How long have             │
│   you had this pain?)                                │
│                                                      │
│   Patient: "کل رات سے"                               │
│   (Since last night)                                 │
│                                                      │
│   Agent: "میں آپ کو ابھی ڈاکٹر علی رضا،              │
│   کارڈیالوجسٹ کے پاس بھیج رہی ہوں۔                   │
│   کمرہ نمبر 302، تیسری منزل۔"                        │
│   (I'm sending you to Dr. Ali Raza,                  │
│   Cardiologist. Room 302, 3rd floor.)                │
│                                                      │
│   Total time: 90 seconds                             │
│                                                      │
└──────────────────────────────────────────────────────┘
```

SLIDE 4: How It Works

The Technology Behind the Voice

The Pipeline (under 5 seconds end-to-end)

```
   Patient Speaks          AI Processes                AI Responds
  ┌────────────┐         ┌──────────────────────┐        ┌────────────┐
  │ Browser    │ WebRTC  │  Cloud AI Pipeline   │ WebRTC │ Browser    │
  │ Mic        │────────→│                      │────────→│ Speaker    │
```



Key Technologies

Component	What It Does	Why It's Special
Deepgram Nova-3 / OpenAI gpt-4o-transcribe	Converts speech to text in real-time	Confidence-filtered — drops background noise (< 0.70). 95%+ accuracy
Silero VAD	Neural voice activity detection	Filters distant speakers (confidence=0.80, min_volume=0.65)
Triage Scoring Engine	Matches symptoms to 70+ conditions	173 aliases, exact matching, cross-condition boost, age/gender rules
GPT-4o-mini	Understands context, generates responses	Follows triage directive — MUST refer to highest-scoring department
OpenAI TTS / ElevenLabs	Converts text to natural speech	Human-like voices, number-to-Urdu conversion, audio caching
WebRTC	Streams audio browser ↔ server	Zero latency, built-in echo cancellation, works through firewalls
Pipecat-AI	Orchestrates the full pipeline	Production-grade, handles interruptions, built-in metrics

What Makes It Different

- **Clinical triage, not chatbot** — Real-time disease cluster scoring against 70+ conditions. Live confidence percentages per department.
- **173 multilingual symptom aliases** — Patient says "کھانسی" → scores as "cough" across all conditions. Urdu, Roman Urdu, English, and common mistranscriptions covered.
- **Intelligent scoring** — Cross-condition boost, vital boost, severity boost, age/gender department rules. Scoring aligns with LLM referral decisions.
- **Two-layer noise filtering** — Silero VAD (neural, 0.80 confidence) + Deepgram STT confidence threshold (0.70). Background speakers filtered without blocking single-word patient responses.
- **Culturally intelligent** — Gender grammar, cultural honorifics, pure Urdu script output (no English mixing). Number-to-Urdu TTS conversion.

- **Hospital-specific** — Configured for specific doctors, rooms, departments. Doctor availability checked before referral.
- **Live triage dashboard** — Urdu tooltips on hover for all 199 symptoms. Real-time scoring updates with each new symptom.
- **265 automated tests** — 188 triage scoring + 77 platform tests, 100% pass rate.

SLIDE 5: The Market

Healthcare AI: Fastest Growing AI Segment

Global Market Opportunity

Market	2024	Projected	CAGR
Global Healthcare AI	$14.9B	$110.6B by 2030	38.6%
Conversational AI in Healthcare	$13.7B	$48.9B by 2030	23.8%
Middle East Healthcare AI	$436M	$8.4B by 2033	37.0%
Middle East Hospital Services	$114B	$172.8B by 2033	4.7%

Sources: Grand View Research, MarketsandMarkets, BCC Research (2024-2025)

Our Target Markets

Phase 1 — Pakistan (Year 1):
- **1,830 hospitals** (1,147 public + 683 private), 123,000+ beds
- $7B hospital market (2024), growing at 4.3% CAGR
- 240M population, 4+ major languages, <5% AI adoption at reception

Phase 2 — Saudi Arabia & Gulf (Year 1-2):
- **Saudi Arabia**: 290+ hospitals + 2,300 primary care centers under privatization. $67B healthcare market. Vision 2030 investing $65B in healthcare infrastructure. Recruiting 175K healthcare professionals by 2030
- **UAE**: 972 hospitals and healthcare facilities. $22B healthcare market, growing to $50B by 2029 (8% CAGR)
- **Qatar**: $12B healthcare sector, ranked #1 in Arab world for healthcare quality
- **Kuwait, Bahrain, Oman**: Mandatory health insurance driving expansion. Active digital health initiatives

Phase 3 — Malaysia & Expansion (Year 2-3):
- **Malaysia**: 355 hospitals (148 public + 207 private). $28B healthcare market, growing to $48B by 2028 (13.5% CAGR). Government MyDigital push
- **India**: 70,000+ hospitals. World's largest healthcare market by volume
- **Indonesia**: 3,000+ hospitals, 280M population

Total Addressable Market (TAM)

Segment	Hospitals	Healthcare Market
Pakistan	~1,830	$7B
Saudi Arabia	~2,590	$67B
UAE	~972	$22B
Other GCC (Qatar, Kuwait, Bahrain, Oman)	~500	$30B
Malaysia	~355	$28B
Phase 1-3 Total	**~6,250**	**$154B**
India + Indonesia (expansion)	~73,000	$200B+

Sources: WHO EMRO, Statista, US-Saudi Business Council, UAE Open Data, Malaysia MOH (2024)

SLIDE 6: Competitive Advantage

Why We Win

1. Real-Time Clinical Triage with Multilingual Voice — Nobody Else Does This

Most healthcare AI is **text-based**, **English-only**, and **diagnosis-focused**. We are the only solution that:
- Performs **real-time disease cluster scoring** against 70+ conditions with live confidence percentages
- Handles **voice conversations** in Urdu, Arabic, and Punjabi (RTL languages) with 173 symptom aliases
- Uses **intelligent scoring** with cross-condition boost, age/gender rules, and exact symptom matching

- Filters **background noise** with two-layer VAD + STT confidence filtering
- Applies **cultural communication norms** (Islamic greetings, gender grammar, pure Urdu script)

2. Hospital-Specific, Not Generic

Generic Health AI	Our AI Clinical Triage System
"You might have condition X"	"REFER TO → Dr. Ali Raza, Cardiology, Room 302 (48% match)"
No real-time scoring	Live disease cluster scoring — 70+ conditions, ICD-10 codes
No context of hospital	Knows every doctor, room, schedule, and availability
Diagnosis-focused (risky)	Triage-focused (safe, referral-based, scoring-driven)
English only	173 symptom aliases across Urdu, English, Roman Urdu
Requires medical certification	Reception function — no medical claims

3. Deployment Speed

```
Generic EHR Integration:  6-12 months, $200K+ implementation
Our Setup:                2-4 hours, $500/month
```

We don't replace the EHR — we sit **in front** of it. The agent handles the reception interaction, then passes the documented conversation to the existing hospital system.

4. Cost Structure

Our per-interaction cost of **$0.04-0.06** (dropping to **$0.02-0.04** with TTS caching at steady state) allows us to price at $0.10-0.25/interaction and maintain **70-85% gross margins** — comparable to SaaS economics, not healthcare services economics. A detailed cost analysis dashboard covers all 24 STT+LLM+TTS combinations with monthly projections.

SLIDE 7: Business Model

Multiple Revenue Streams, SaaS Economics

Pricing Tiers

Tier	Price	Includes	Target
Starter	$500/month	2,000 interactions, 1 language, basic personas	Small clinics
Professional	$1,500/month	10,000 interactions, 3 languages, all personas, analytics	Medium hospitals
Enterprise	$3,000/month	Unlimited interactions, all languages, custom personas, EMR integration, SLA	Hospital chains
Kiosk	$5,000 setup + $200/month	Hardware + software, self-service terminal	Reception replacement

Unit Economics

Metric	Value
Average Revenue Per Hospital	$1,500/month ($18K/year)
Cost per Interaction	$0.04-0.06
Average Interactions per Hospital	3,000/month
Cost per Hospital	$150-180/month
Gross Margin	**88-90%**
Customer Acquisition Cost (CAC)	$2,000-5,000
Lifetime Value (LTV)	$54K (3-year retention)
LTV/CAC Ratio	**11-27x**

Path to Revenue

Quarter	Milestone	Revenue
Q2 2026	5 pilot hospitals (Pakistan)	$7.5K/month

Quarter	Milestone	Revenue
Q4 2026	25 hospitals (Pakistan + UAE)	$37.5K/month
Q2 2027	100 hospitals (3 countries)	$150K/month
Q4 2027	300 hospitals (5 countries)	$450K/month
Q4 2028	**1,000 hospitals**	**$1.5M/month ($18M ARR)**

SLIDE 8: Traction & Proof Points

What We've Built and Proven

Technical Milestones (Completed)

Milestone	Status	Details
Core voice pipeline	Done	STT → LLM → TTS with <5s latency
6 language support	Done	Urdu, English, Hindi, Arabic, Punjabi, Malay
RTL/LTR auto-switching	Done	UI adapts direction per language
10 configurable personas	Done	5M/5F with gender-appropriate grammar
12 specialty triage	Done	Full doctor panel with real-time availability
WebRTC audio	Done	Browser-based, zero installation
Session persistence	Done	Settings survive refresh, conversations logged
Settings lock during calls	Done	Prevents config changes during active sessions
Automated test suite	Done	77 tests, 100% pass rate
Performance benchmarks	Done	All metrics under threshold
Dual STT providers	Done	Deepgram Nova-3 + OpenAI gpt-4o-transcribe, switchable from UI
TTS audio caching	Done	Disk-backed LRU (500 entries), ~85% cost savings at steady state
Patient screening card	Done	Real-time data extraction with printable triage slip
Symptom-to-vitals mapping	Done	15-category intelligent vitals collection

Milestone	Status	Details
Conversation flow stepper	Done	Visual progress: Name → Symptoms → Duration → Vitals → Refer
Live model switching	Done	Change LLM/TTS voice mid-call; STT changes on next session
Urdu script output	Done	Proper اردو رسم الخط instead of Roman Urdu
Cost analysis dashboard	Done	24 combinations with monthly projections
Triage scoring engine	Done	70+ conditions, 12 departments, real-time scoring
173 symptom aliases	Done	Urdu + English + Roman Urdu + mistranscriptions
Two-layer noise filtering	Done	Silero VAD (0.80) + Deepgram confidence (0.70)
Age/gender department rules	Done	Pediatrics age-gated, Gynecology gender-gated
Cross-condition boost	Done	+15% per additional strong condition in department
Exact symptom matching	Done	"cough" ≠ "persistent cough" (prevents cross-contamination)
Doctor availability check	Done	Unavailable doctors skipped in referral
Urdu triage tooltips	Done	All 199 symptoms with Urdu hover translations
Off-topic answer handling	Done	Re-asks unanswered symptom questions
Number-to-Urdu TTS	Done	Room "302" spoken as "تین سو دو"
Pure Urdu output	Done	No English word mixing in responses
188 triage scoring tests	Done	Scoring accuracy, alias expansion, referral scenarios

Key Performance Indicators

KPI	Target	Actual
End-to-end voice latency	< 5 seconds	3-5 seconds
STT accuracy	> 90%	95%+
Triage test pass rate	100%	100% (188/188)
Platform test pass rate	100%	100% (77/77)
Symptom alias coverage	100%	100% (199/199 symptoms covered)
Page load time	< 200ms	< 50ms

KPI	Target	Actual
Languages supported	4+	6

SLIDE 9: The Team

Experienced AI + Healthcare Engineering

Leadership

Dr. Aqib Mumtaz, Ph.D. — Head of AI
- Ph.D. in Computer Science (AI specialization), COMSATS University, Lahore (2019-2024)
- Research collaboration with University of Bergen, Norway — Anomaly Detection
- 19+ years experience in AI/ML, Deep Learning, Computer Vision, NLP
- Published researcher — 5 peer-reviewed papers in anomaly detection, video surveillance, deep learning
- Expertise: ASR (OpenAI Whisper, Wave2Vec2), Transformer architectures, attention mechanisms, real-time AI systems
- State-of-the-art AI stack: GPT-4o, Qwen3-ASR, Deepgram Nova-3, ElevenLabs neural TTS, WebRTC streaming, pipecat-ai orchestration
- Previously led AI research teams of 22+ engineers (SlashNext — phishing detection AI)
- Reviewed Pakistan's National AI Policy (ITU collaboration)
- Award and Honour — University of Kagoshima, Japan

Core Competencies

Capability	Evidence
ASR/NLP	Production ASR system with 70,000+ ICD-10 codes, morphological indexing, fuzzy matching
Deep Learning	Transformer architectures, attention mechanisms, Whisper fine-tuning, generative models
Real-time Systems	WebRTC-based transcription bots running in production
Medical AI	Symptom extraction, negation detection, drug/lab entity recognition
Multilingual	Urdu, English, Hindi, Arabic, Punjabi, Malay — with cultural adaptation
Research	5 published papers, international collaborations (Norway, Japan)

Capability	Evidence
Full-Stack	Frontend, backend, infrastructure, cloud deployment — end-to-end delivery

Hiring Plan (with funding)

Role	Count	Timeline
AI/ML Engineer	2	Immediate
Full-Stack Developer	1	Month 1-2
Healthcare Domain Expert	1	Month 2-3
Business Development (MENA)	1	Month 1
DevOps / SRE	1	Month 3

SLIDE 10: What We're Asking For

Seed Round: $500K - $1M

Use of Funds

Category	Allocation	Purpose
Engineering (50%)	$250-500K	4-5 engineers for 12 months. EMR integration, analytics dashboard, kiosk hardware, phone integration
Go-to-Market (25%)	$125-250K	5 pilot deployments, sales team for Pakistan/UAE, marketing materials, conference presence
Infrastructure (15%)	$75-150K	Cloud hosting, TURN servers, API costs for pilot hospitals
Operations (10%)	$50-100K	Legal (healthcare compliance), accounting, office

Milestones This Round Achieves

Timeline	Milestone	Metric
Month 3	5 pilot hospitals live	5 paying customers
Month 6	EMR integration (HL7/FHIR)	Enterprise feature

Timeline	Milestone	Metric
Month 9	25 hospitals across 2 countries	$37.5K MRR
Month 12	Series A ready	$50K+ MRR, 50+ hospitals

What Investors Get

- **Equity**: 15-20% (seed stage)
- **Market**: First-mover in RTL voice AI for healthcare
- **Technology**: Production-proven, 77-test-validated platform
- **Revenue path**: Clear unit economics with 88-90% gross margins
- **Expansion**: Each new language/country multiplies TAM without rebuilding core tech

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SLIDE 11: Why Now?

The Convergence of Five Trends

1. AI Quality Just Crossed the Threshold

GPT-4o-mini delivers human-level conversation quality at $0.01/interaction. Two years ago, this cost $1.00+ and was too slow for real-time voice.

2. Voice AI Infrastructure Matured

Deepgram, ElevenLabs, and Pipecat have made production-grade voice pipelines possible in weeks, not years. WebRTC handles browser audio natively.

3. Post-COVID Healthcare Digitalization

Hospitals are actively seeking automation. COVID proved that human-only reception is a single point of failure. Government healthcare digitalization programs are funded and active (Saudi Vision 2030, Malaysia MyDigital, Pakistan Digital Health).

4. Multilingual AI Just Became Viable

Until 2024, AI voice quality in Urdu, Arabic, and Hindi was poor. Nova-3 and multilingual TTS models now deliver production-quality in these languages for the first time.

5. Healthcare Staffing Crisis

Global healthcare worker shortage of 14.7 million (WHO, 2023), projected 11M by 2030. Reception staff turnover is 30-40% annually. AI doesn't call in sick, doesn't need training, and works 24/7.

SLIDE 12: The Ask

Let's Transform Healthcare Together

What We Offer

- **Working product** — Not a concept. A production-ready AI clinical triage system you can demo today.
- **Real-time scoring** — 70+ conditions, 12 departments, live confidence percentages. 188 triage tests, 100% pass rate.
- **173 multilingual aliases** — Urdu, Roman Urdu, English, and common STT mistranscriptions. All 199 symptoms with Urdu tooltips.
- **Intelligent noise filtering** — Two-layer: Silero VAD (neural) + Deepgram confidence. Works in noisy hospital environments.
- **Scoring-LLM alignment** — Triage engine decides the doctor, LLM follows the directive. No conflicting recommendations.
- **Cultural intelligence** — Pure Urdu script, gender grammar, Islamic greetings, number-to-Urdu TTS.
- **Hospital-specific** — Configured for real doctors, rooms, schedules, and availability.
- **Cost-optimized** — TTS caching ~85%, dual STT providers, 265 automated tests.
- **Massive market** — 6,250+ hospitals across Pakistan, Saudi Arabia, UAE, Gulf, and Malaysia.

What We Need

- **$500K-1M seed funding** to deploy in 25 hospitals and build enterprise features
- **Strategic partners** in healthcare (hospital chains, health ministries, healthcare IT vendors)
- **Advisors** with healthcare operations and MENA/South Asia business experience

The Opportunity

The first company to own **voice AI at hospital reception** in South Asia and MENA will have an unassailable position. Every hospital is a potential customer. Every patient interaction builds our data moat. Every language we add multiplies our market.

We've built the technology. Now we need to deploy it.

Contact

AI-Powered Healthcare Solutions

For demo, or inquiries:
- Email: aqib.mumtaz@gmail.com

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